Computational Neuroscience — Module 02: Agents — Study Questions

1. What is the cortical hierarchy? How is information organized from lower to higher areas?
2. What is the role of the basal ganglia in Active Inference?
3. How does dopamine function as a precision signal? What does it tell the brain?
4. What happens to action selection when dopamine is depleted (as in Parkinson's disease)?
5. What is the thalamus and what role does it play in precision gating?
6. How does cortical feedback to the thalamus implement attention?
7. What is a forward model? Which brain structure specializes in forward models?
8. Why does cerebellar damage cause clumsy movement? Explain using prediction error.
9. What is the role of acetylcholine in Active Inference?
10. Compare the roles of dopamine and acetylcholine in precision weighting.
11. What does noradrenaline do in the context of Active Inference?
12. How might serotonin relate to patience and temporal planning?
13. How does the cortical hierarchy implement a generative model?
14. What flows top-down in the cortical hierarchy? What flows bottom-up?
15. How do the basal ganglia evaluate which action to take?
16. Draw or describe the basic circuit: cortex → basal ganglia → thalamus → cortex.
17. Why is the thalamus sometimes called a "relay station"? Is this description complete?
18. How does the cerebellum help you catch a ball?
19. What is the difference between the basal ganglia's role and the cerebellum's role in action?
20. How does the computational neuroscience view of agents differ from the cognitive science view (Unit 1, Module 02)?